Daily Geometry Drill #2

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Circle Equations · Tangents · Line–Circle Tests · Chord Lengths.

Section A Rapid Recognition

1. Write down the centre of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.

Ans: "$(3, 4)$"

2. For which value of k does $x^2 + y^2 - 6x + 4y + k = 0$ represent a circle of radius 5?

Ans: -12

3. Write down the equation of the circle with centre $(2, -1)$ and radius 5.

Ans: "$x^2 + y^2 - 4x + 2y - 20 = 0$"

4. Write down the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.

Ans: 5

5. The point $(x, 4)$ lies on the circle $x^2 + y^2 = 25$ with $x > 0$. Find x.

Ans: 3

6. The radius from the origin to the point $(3, 4)$ on $x^2 + y^2 = 25$ has what gradient?

Ans: $\frac{4}{3}$

7. The tangent to $x^2 + y^2 = 25$ at the point $(3, 4)$ has what gradient?

Ans: $-\frac{3}{4}$

8. Write down the equation of the circle with centre $(3, -4)$ that passes through the origin.

Ans: $(x-3)^2 + (y+4)^2 = 25$

9. The centre of $x^2 + y^2 - 6x - 8y + 24 = 0$ is $(3, 4)$. Find the perpendicular distance of this centre from the line $4x + 3y - 9 = 0$.

Ans: 3

10. A chord of the circle $x^2 + y^2 = 25$ is at perpendicular distance 3 from the origin. Find the length of the chord.

Ans: 8

Section B Manipulation Drills

1. Which of the following is the equation of the circle with centre $(3, -2)$ and radius 4?

A) $x^2 + y^2 - 6x + 4y - 3 = 0$

B) $x^2 + y^2 - 6x + 4y + 9 = 0$

C) $x^2 + y^2 + 6x - 4y - 3 = 0$

D) $x^2 + y^2 - 6x - 4y - 3 = 0$

Ans: A) $x^2 + y^2 - 6x + 4y - 3 = 0$

2. The circle $x^2 + y^2 + 2x - 4y - 11 = 0$ has what radius?

A) 2

B) 3

C) 4

D) $\sqrt{5}$

Ans: C) 4

3. The point $(1, 2)$ lies on the circle $x^2 + y^2 = 5$. The equation of the tangent to the circle at $(1, 2)$ is?

A) $y = -\frac{1}{2}x + \frac{5}{2}$

B) $y = 2x$

C) $y = -2x + 5$

D) $y = \frac{1}{2}x$

Ans: A) $y = -\frac{1}{2}x + \frac{5}{2}$

4. A circle of form $x^2 + y^2 + 2gx + 2fy + c = 0$ passes through the origin. Deduce c.

A) $c = 1$

B) $c = 0$

C) $c = g + f$

D) $c = -1$

Ans: B) $c = 0$

5. The perpendicular distance from the centre of $x^2 + y^2 = 25$ to the line $y = 2x + 3$ is?

A) $\dfrac{3}{\sqrt{5}}$

B) $\dfrac{3}{5}$

C) 3

D) 2

Ans: A) $\frac{3}{\sqrt{5}}$

6. Which of the following is a tangent to the circle $x^2 + y^2 = 4$?

A) $y = x + 2\sqrt{2}$

B) $y = x + 4$

C) $y = 2x + 2$

D) $y = 2x + 4$

Ans: A) $y = x + 2\sqrt{2}$

7. A chord of the circle $x^2 + y^2 = 13$ has length $4\sqrt{3}$. What is the perpendicular distance of the chord from the centre?

A) 3

B) 2

C) $\sqrt{5}$

D) 1

Ans: D) 1

8. The line $y = mx + 4$ is a tangent to the circle $x^2 + y^2 = 4$. Find the two possible values of m.

A) ± 1

B) $\pm\sqrt{3}$

C) ± 2

D) $\pm\sqrt{2}$

Ans: B) $\pm\sqrt{3}$

9. The circle with centre $(6, a)$ and radius 6 is tangent to the y-axis and passes through the point $(0, 8)$. Find a.

A) 1

B) 3

C) 5

D) 8

Ans: D) $a = 8$

10. Write down the length of the chord cut from the line $y = x + 1$ by the circle $x^2 + y^2 = 25$.

Ans: $7\sqrt{2}$

Section C Substitution & Structure (≤90 s each)

Harder: hidden tangency, chord bisection, and the distance-vs-discriminant decision. No brute substitution.

1. A tangent to the circle $x^2 + y^2 = r^2$ meets the positive x- and y-axes at A and B. The triangle OAB has area 150 and $AB = 25$. What is r? *(after TMUA 2017 Paper 1 Q6)*

A) 6

B) 12

C) 15

D) $\frac{25}{2}$

Ans: B) 12

2. The point $P(8, 6)$ is outside the circle $x^2 + y^2 = 36$. Tangents from P meet the circle at A and B. What is the length AB?

A) $\frac{12\sqrt{13}}{13}$

B) $\frac{24\sqrt{13}}{13}$

C) $\frac{48}{5}$

D) $\frac{72}{5}$

Ans: C) $\frac{48}{5}$

3. The point Q moves on the circle $x^2 + y^2 - 2x + 4y - 20 = 0$, and P is the point $(13, 3)$. What is the least possible distance PQ?

A) 13

B) 18

C) 12

D) 8

Ans: D) 8

4. A regular hexagon of area $54\sqrt{3}$ has all six vertices on the circle $x^2 + y^2 - 18x - 22y + c = 0$. What is c? *(after TMUA 2017 Paper 1 Q9)*

A) 196

B) 166

C) 130

D) 238

Ans: B) 166

5. The equation $x^2 - 2px + y^2 - 4y - p^2 + 6p + 4 = 0$ describes a circle that touches the y-axis. What is its radius? *(after TMUA 2022 Paper 1 Q2)*

A) 6

B) 3

C) 2

D) 0 or 6

Ans: A) 6

6. The circle $x^2 + y^2 - 4x + 2y - 11 = 0$ and the line $x - y + 1 = 0$: determine the number of intersection points without solving the system.

A) 0 (line misses)

B) 1 (tangent)

C) 2 (secant)

D) It cannot be decided without solving the system

Ans: C) 2 (secant)

7. The circle $x^2 + y^2 - 8x - 4y - 21 = 0$ is cut by the y-axis. What is the length of the chord intercepted on the y-axis?

A) $2\sqrt{21}$

B) $2\sqrt{41}$

C) 8

D) 10

Ans: D) 10

8. The line $y = kx + 5$ is tangent to the circle $x^2 + y^2 - 6x - 8y + 24 = 0$. How many distinct values of k satisfy this?

A) 0

B) 1

C) 2

D) Infinitely many

Ans: C) 2

Section D Challenge (TMUA / SMC / BMO1 difficulty)

Hardest 5: distance invariants, hidden right angles, and a construction you must spot.

1. The circle $C_1 : (x+2)^2 + (y-3)^2 = 18$ and a circle C_2 with centre $(7, -6)$ and radius r touch each other. What is the sum of the possible values of r? *(after TMUA 2018 Paper 1 Q3)*

A) $6\sqrt{2}$

B) $12\sqrt{2}$

C) $18\sqrt{2}$

D) $9\sqrt{2}$

Ans: C) $18\sqrt{2}$

2. A circle is translated 3 units in the negative x-direction, then reflected in the x-axis, then enlarged by scale factor 4 about the *origin*. The final circle is $(x-8)^2+(y+16)^2=80$. What was the centre of the original circle? *(after TMUA Specimen Paper 1 Q9)*

A) $(-1, 4)$

B) $(5, 4)$

C) $(5, -4)$

D) $(\frac{11}{4}, 4)$

Ans: B) $(5, 4)$

3. Points P and Q lie on a circle with centre O and radius 6. The point R moves on the major arc PQ, and the greatest possible area of triangle PRQ is $27\sqrt{3}$. What is $\angle POQ$? *(after TMUA 2022 Paper 1 Q14)*

A) $90°$

B) $120°$

C) $60°$

D) $150°$

Ans: B) $120°$

4. Two tangents are drawn from the origin to the circle $(x-5)^2+(y-5)^2=5$. What is the tangent of the acute angle between them?

A) $\frac{3}{4}$

B) $\frac{4}{3}$

C) $\frac{5}{2}$

D) 1

Ans: A) $\frac{3}{4}$

5. The circle through $(0,0)$, $(8,0)$ and $(2,6)$ meets the y-axis again at Q. How far is Q from the origin?

A) $2\sqrt{5}$

B) 6

C) 2

D) 4

Ans: D) 4